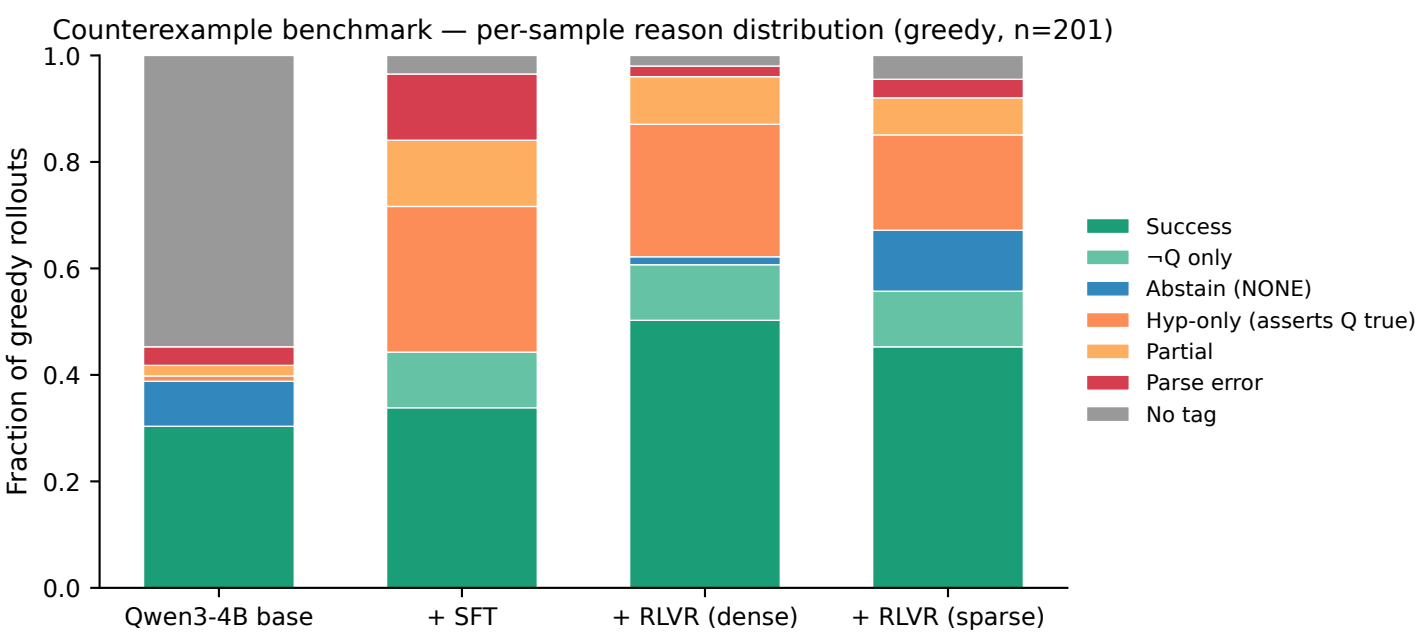
Counterexample benchmark — per-sample reason distribution (greedy, n=201)
Fraction of greedy rollouts
1.0
0.8
0.6
0.4
0.2
0.0
Qwen3-4B base
+ SFT
+ RLVR (dense)
+ RLVR (sparse)
Success
¬Q only
Abstain (NONE)
Hyp-only (asserts Q true)
Partial
Parse error
No tag